

Tanmay Tomar

📍 Alwar, Rajasthan ✉ tomartanmay1109@gmail.com 📞 +91-78508-24437
🐙 github.com/TANMaYtO 🔗 linkedin.com/in/tanmay-tomar-430500281 🌐 tanmaytomar.vercel.app

Achievements

- **RBIH 2026 National Finalist (Top 4 / 25,000+ teams)** — Sole architect of a real-time fraud detection pipeline; defended system design before IISc professors and senior industry data scientists at the national finals.
- **Amazon ML Challenge 2025** — Ranked in the **top 3%** of 25,000+ global participants.

Technical Skills

Agentic & LLM Systems: LangGraph, Google Gemini API, Large Language Models (LLMs), Multi-Agent Systems, Kokoro-ONNX TTS, FFmpeg

ML & Deep Learning: PyTorch, Hugging Face Transformers, Sentence Transformers, RLHF, RAG, FAISS, Vector Search, Diffusion Models

Data & Predictive Modeling: Polars, Pandas, CatBoost, LightGBM, XGBoost, Bayesian Calibration

Web & Infrastructure: FastAPI, Next.js 14, Docker, Supabase, Cloudflare R2, Stripe, AWS EC2, tree-sitter

Languages: Python, SQL, TypeScript

Projects

Cronus — Autonomous YouTube Shorts SaaS *LangGraph · Gemini · Kokoro-ONNX · FFmpeg · Next.js 14 · Stripe*
github.com/TANMaYtO/YT-AUTONOMOUS-BOT *Apr 2026 – Jun 2026*

- Bootstrapped and shipped a multi-tenant YouTube Shorts automation SaaS to **20+ daily active users and 10+ paying Pro subscribers**; 7-node LangGraph pipeline drives trend discovery, Gemini Flash scriptwriting, TTS, 1080×1920 FFmpeg assembly, and YouTube Data API v3 upload with zero human intervention.
- Eliminated cloud TTS costs entirely by running Kokoro-ONNX locally at **sub-2s inference per segment**; 2-pass FFmpeg engine produces frame-accurate, word-level kinetic subtitles from per-token timestamps compiled as **.ass** overlays onto H.264 vertical video.
- Enforced complete multi-tenant isolation via Supabase Row Level Security and **AES-256-GCM encryption** of YouTube OAuth 2.0 refresh tokens; stateless Cloudflare R2 eliminates all worker disk state, with Stripe gating Free/Pro feature access.

PRISM — Autonomous PR Review System *LangGraph · Gemini · FAISS · FastAPI · Next.js · Supabase*
github.com/TANMaYtO/PRISM *Jun 2026*

- Built a parallel 4-agent PR review system on LangGraph (bug detector, security scanner, logic auditor, style checker) executing concurrently via graph fan-out and streaming findings in real time over SSE to a Next.js dashboard; handles diffs exceeding 1,200 changed lines with all reviews persisted to Supabase.
- Engineered an AST-based Code Knowledge Graph via tree-sitter and Python **ast**, extracting **6,000+ function definitions and 29,000+ call edges** as structural RAG context (callee/caller chains, class hierarchies, import graphs) in **under 5s** — zero LLM extraction overhead, deterministic, outperforming community-summary GraphRAG for code-search precision.
- Benchmarked against human maintainers across **19 real langchain-ai/langgraph PRs**, achieving **0.33 F1 / 0.48 recall**; surfaced **103 issues vs 83 human comments**, including an async event-loop deadlock and a **TypeError** in LC:2 deserialization — both merged undetected to production.

V24 Fraud Detection — RBIH 2026 National Finalist *Polars · CatBoost · LightGBM · Bayesian Calibration*
Nov 2025 – Feb 2026

- Processed large-scale financial transaction datasets via an out-of-core Polars pipeline, engineering **sub-second temporal variance features** to fingerprint automated bot networks and suppress decoy cluster interference across spatial fraud distributions.
- Trained a CatBoost + LightGBM ensemble with Bayesian post-hoc calibration, achieving **0.910 F1** on a held-out test set with calibrated probabilistic cluster scores and **sub-minute real-time scoring latency** under the live evaluation load at RBIH 2026 national finals.

Professional Experience

Innodata Inc. *Jan 2026 – Mar 2026*
AI Data Associate — Contract, Remote

- Evaluated **500+ prompts weekly** on proprietary multimodal GenAI models, identifying instruction-following failures, temporal consistency gaps, and physics-based hallucination modes as structured RLHF fine-tuning signal.
- Maintained **95%+ QA accuracy** across model-generated video outputs; surfaced failure patterns into alignment and safety fine-tuning datasets, contributing to measurable reduction in hallucination rates across successive model versions.

Education

MITRC, Alwar *Expected May 2027*
B.Tech — Artificial Intelligence & Data Science